

PLASTIC DETECTION USING YOLO

Plastic pollution has emerged as one of the most critical environmental challenges, significantly impacting marine ecosystems, biodiversity, and long-term ecological balance. One of the major difficulties in addressing this issue is the lack of efficient automated systems capable of accurately detecting and localizing plastic waste in real-world environments, particularly in underwater and marine settings where visibility and object complexity are high.

To address this problem, we propose an intelligent system titled “**Plastic Detection using YOLO**”, which leverages deep learning-based object detection techniques to identify and localize marine plastic debris in images. Unlike traditional classification approaches, the system uses a YOLO (You Only Look Once) model to perform real-time object detection, generating bounding boxes around detected debris along with confidence scores indicating prediction reliability.

The model is trained on the **TrashCan dataset**, which contains annotated underwater images of marine waste. Given an input image through a Streamlit-based web interface, the system processes it using the trained YOLO network and outputs the locations of plastic debris within the scene. Each detection is visually represented using bounding boxes, enabling precise spatial understanding of waste distribution.

This approach enhances automation in environmental monitoring by enabling both detection and localization of plastic waste, making it more suitable for real-world deployment scenarios such as autonomous ocean-cleaning systems and robotic inspection platforms. Ultimately, the proposed system contributes toward scalable marine pollution monitoring and supports technological efforts aimed at preserving aquatic ecosystems.

TEAM NUMBER: 10

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